

HARBIN, China

WMO: 509530

Lat: 45.9340N

Lon: 126.5810E

Elev: 386

StdP: 14.49

Time Zone: 8.00 (E08)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-17.3	-13.3	-23.7	1.5	-15.4	-20.0	1.9	-11.6	16.0	14.3	14.2	12.7	3.1	180	0.503

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	14.0	88.8	69.0	86.2	70.0	83.8	69.6	76.0	82.9	74.5	81.2	73.0	79.6	8.0	220

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
73.8	127.9	79.9	72.3	121.4	78.7	70.8	115.2	77.5	39.9	83.2	38.4	81.5	37.1	79.8	
															(p)
															82.9

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature								
1%		2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
Min		Max		Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)		(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
16.8	14.6	12.9	DB	-21.2	94.1	5.8	3.2	-25.4	96.4	-28.7	98.3	-32.0	100.1	-36.1	102.4
			WB	-21.6	78.0	5.6	2.1	-25.6	79.5	-28.9	80.7	-32.1	81.9	-36.2	83.4

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	41.5	1.2	10.6	27.2	45.9	59.8	70.7	74.7	71.5	60.5	44.5	23.6	6.0
	DBStd	27.01	8.16	9.94	10.51	8.90	7.57	6.12	4.12	4.68	6.47	8.57	11.15	8.91
	HDD50	5890	1514	1102	708	182	8	0	0	0	4	214	794	1364
	HDD65	9345	1979	1522	1173	574	200	18	1	7	163	636	1243	1829
	CDD50	2785	0	0	1	60	311	621	765	665	318	43	1	0
	CDD65	763	0	0	0	2	37	189	301	207	27	0	0	0
	CDH74	5927	0	0	0	32	437	1640	2284	1333	197	4	0	0
	CDH80	1626	0	0	0	4	125	562	640	275	20	0	0	0
Wind	WSAvg	5.9	5.0	5.6	6.8	7.5	7.3	6.1	5.4	5.1	5.2	5.8	6.1	5.2
Precipitation	PrecAvg	20.7	0.2	0.2	0.5	0.8	2.0	3.8	5.5	4.1	1.9	1.0	0.6	0.3
	PrecMax	32.4	0.5	0.7	1.5	2.4	4.4	8.2	14.1	9.4	3.4	2.2	1.8	0.8
	PrecMin	15.1	0.0	0.0	0.0	0.0	0.4	0.6	1.8	1.3	0.2	0.4	0.0	0.0
	PrecStd	3.9	0.1	0.2	0.4	0.6	1.1	1.8	2.3	2.0	0.9	0.5	0.4	0.2
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	29.1	41.4	60.8	78.3	88.8	94.8	91.8	88.0	82.7	73.1	56.1	38.4
		MCWB	25.0	34.1	45.6	54.0	61.0	66.5	72.5	74.7	68.1	58.4	47.3	35.3
	2%	DB	22.4	36.3	52.9	72.0	83.5	89.5	88.3	85.4	79.1	67.4	50.1	28.9
		MCWB	19.4	30.2	41.0	51.7	59.9	66.0	72.9	73.0	65.5	55.0	42.4	25.9
	5%	DB	18.1	31.6	47.5	67.2	78.9	86.3	85.9	83.1	76.0	62.6	44.9	24.0
		MCWB	15.6	26.7	37.2	49.5	58.1	66.5	72.7	71.8	63.3	51.6	38.7	21.3
	10%	DB	14.0	26.5	43.0	62.3	74.3	82.8	83.6	80.9	72.9	58.4	39.3	19.7
		MCWB	11.9	22.5	34.5	47.0	56.3	65.3	72.3	70.4	61.2	49.3	34.3	17.4
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	25.3	34.7	46.2	56.6	65.5	73.5	78.9	77.7	71.5	60.1	48.6	34.9
		MCDB	28.6	40.1	58.2	72.5	80.4	83.4	84.8	84.7	78.3	69.7	54.6	38.0
	2%	WB	19.5	30.9	41.5	53.3	62.5	71.3	76.6	75.5	68.4	56.4	42.9	26.0
		MCDB	22.2	36.0	51.2	68.3	75.8	81.5	83.4	82.4	75.7	64.8	48.6	28.4
	5%	WB	15.6	26.5	38.0	50.3	60.1	69.5	75.3	74.0	65.9	53.4	38.7	21.4
		MCDB	18.0	31.7	46.4	64.9	73.9	79.2	82.1	80.2	72.9	60.6	44.6	23.9
	10%	WB	12.0	22.6	35.0	47.7	58.0	68.0	73.9	72.5	63.4	50.2	34.6	17.4
		MCDB	13.9	26.4	42.5	60.8	71.5	77.8	80.7	78.3	70.2	57.6	39.0	19.7
Mean Daily Temperature Range		MDBR	17.0	19.1	17.8	19.4	19.1	17.1	14.0	14.9	19.3	18.1	14.8	15.1
	5% DB	MCDBR	20.6	21.0	23.4	26.2	26.8	22.2	17.6	17.7	22.7	23.6	20.4	18.8
		MCWBR	18.2	16.9	14.5	13.7	10.9	8.0	6.7	7.6	10.5	13.1	14.7	16.8
	5% WB	MCDBR	20.6	21.0	22.1	23.4	21.1	16.4	14.0	14.3	18.5	20.7	19.5	18.9
		MCWBR	18.4	17.0	14.4	13.5	9.9	7.0	7.1	7.4	10.3	13.1	15.3	17.1
Clear-Sky Solar Irradiance	taub		0.326	0.342	0.416	0.518	0.580	0.620	0.584	0.514	0.461	0.455	0.373	0.335
	taud		2.077	2.057	1.941	1.780	1.707	1.721	1.835	1.991	2.037	1.935	2.058	2.064
	Ebn at Noon		232	256	251	235	226	217	224	235	235	210	210	211
	Edh at Noon		31	39	51	65	73	72	64	52	46	44	32	29
All-Sky Solar Radiation	RadAvg		664	1025	1370	1556	1736	1786	1636	1527	1341	935	614	522
	RadStd		62	69	105	114	160	187	158	112	106	62	48	32

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (6 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page